

Humanware and IT Governance — Detailed Lecture Notes

JKSSB Computer Awareness

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47.1 What humanware means

A computer system is not only metal and programs. It also needs people. The human element of a computer system is called **humanware**, and it is also known as **liveware** or **peopleware**. Humanware means the people who design, build, operate, manage and use computers. It is *neither hardware nor software*; it is the people component of the system.

Definition (Humanware). *The people involved with a computer system — its users, operators, administrators, programmers and support staff. Also called liveware or peopleware.*

Term	Meaning in this lesson
Hardware	The physical parts of the computer.
Software	The programs and instructions the computer runs.
Humanware	The people who use and manage the hardware and software.
User	The person who uses the finished system to do a task.
Administrator	The person who manages the system, network or database.

The rest of this lesson uses these terms in one consistent sense. **Hardware** is the machinery. **Software** is the set of instructions. **Humanware** is the people.

47.2 The people in a computer system

Humanware is not a single role. Many different people take part, and each has a distinct job. The main categories are the following.

- **User (or end user)** — the person who uses the finished system to perform a task. A clerk entering data, a student reading a portal and a customer paying a bill are all users.
- **Operator** — the person who runs the computer system, feeds jobs into it and monitors routine operation.
- **Administrator** — the person who manages the system. A **system administrator** manages servers, user accounts and access rights; a **network administrator** manages the network; a **database administrator** manages the database.
- **Programmer (developer)** — the person who writes, tests and maintains the software.
- **IT support (help desk) and technician** — the people who answer user problems, troubleshoot faults, and install, repair and maintain the hardware.

The distinction to hold is simple. Administrators *manage* the system, operators *run* it, and users *use* its output.

47.3 Hardware, software and humanware

A working computer system needs all three components. Hardware is the physical machine. Software is the set of programs that tell the hardware what to do. Humanware is the people who design, run and use both. Hardware without software does nothing, and both are useless without the people who operate them. This is why humanware is taught as a full component, not as an afterthought.

Example 47.3.1. *A government office installs new computers (hardware) and an accounting program (software). The system still does not work by itself. A programmer must maintain the program, a system administrator must create user accounts, an operator must run the daily jobs, and the office staff (users) must enter the data. Those people are the humanware.*

47.4 The role of IT in governance

Governance is the process of decision-making and its implementation in an organisation or a state. **IT** (Information Technology) and **ICT** (Information and Communication Technology) let government carry out that work over computers and networks. The role of IT in governance is to make government faster, cheaper, more transparent and more accessible to the people it serves.

The main benefits of using IT in governance are the following.

- **Transparency** — citizens can see the rules, the status of their applications and the decisions taken.
- **Accountability** — actions are recorded and can be traced.
- **Efficiency** — less paperwork and faster processing.
- **Convenience** — services are available from home and around the clock.
- **Lower cost and reduced corruption**, because manual handling and physical counters are replaced by digital processes.

The overall aim is **citizen-centric** service: the citizen is placed at the centre of every process.

47.5 IT governance versus e-governance

These two terms are a common near-neighbour pair, and the difference matters.

- **IT governance** is the process by which an organisation directs and controls its own IT resources so that they support its goals. It is *governance of IT*.
- **E-governance** is the use of IT by government to govern and deliver public services. It is *IT in governance*.

The trap is to swap them. IT governance is about managing IT well inside an organisation; e-governance is about using IT to govern the state and serve citizens.

47.6 E-governance basics

E-governance is the application of ICT by government to deliver services, share information and transact with citizens, businesses and other government agencies. It uses the internet, databases and digital records in place of paper and physical counters. It makes government services faster, wider-reaching and easier to audit.

Definition (E-governance). *The use of information and communication technology to improve the way government serves and interacts with the public.*

E-governance is usually described by the parties who interact. The four standard types are the following.

Type	Full form	Interaction
G2C	Government to Citizen	Services to the general public.
G2G	Government to Government	Between government departments.
G2B	Government to Business	With companies and traders.
G2E	Government to Employee	With government staff.

This lesson keeps the treatment at awareness level. Named platforms and detailed G2x matching belong to Lesson 57.

47.7 Digital public services

A **digital public service** is a government service that a citizen can obtain online. Common examples are certificates, licences, land records, bill payment, applications and grievance redressal. Citizens reach these services through service portals and through **Common Service Centres (CSCs)**, which provide a physical access point where help is needed. Typical features are online application, status tracking, digital payment and downloadable documents. A service counts as digital when the citizen can complete the task without visiting an office in person.

47.8 Electronic records

An **electronic record** is information created, stored and retrieved in electronic (digital) form. Examples include e-documents, e-files, scanned papers, database entries and email. Electronic records are easy to store, search, retrieve, share and back up, and they replace large volumes of paper. An electronic record can have legal validity when it is authenticated, for example with a digital signature.

Definition (Electronic record). *Information that is created, stored and retrieved in electronic form, such as an e-document, an e-file or a database entry.*

47.9 Authentication

Authentication is the process of verifying that a user, device or system is who or what it claims to be. It answers the question: “Who are you?” Common methods are a password, a PIN, an **OTP** (One-Time Password), a biometric such as a fingerprint or iris, and a digital certificate. Authentication must succeed before the system grants access.

Definition (Authentication). *The process of confirming a claimed identity before access to a system is allowed.*

Authentication is often confused with **authorization**. **Authorization** is the process of granting or denying permission — it decides what a verified identity is allowed to do. Authentication proves identity; authorization grants or refuses access rights. Authentication always comes first.

Authentication relies on *factors*. There are three standard types: something you **know** (a password or PIN), something you **have** (an OTP on your phone, a token or a smart card), and something you **are** (a fingerprint, iris or face). When two different factors are combined, it is called **two-factor authentication (2FA)**; when two or more are used, it is **multi-factor authentication (MFA)**. Two passwords are not two factors, because they are the same factor used twice.

Example 47.9.1. *A candidate logs in to a government portal with a password and is then asked for an OTP sent to the registered mobile number. The password is something the candidate knows; the OTP is something the candidate has. Together they form two-factor authentication. The portal then decides which forms the candidate may open — that decision is authorization, not authentication.*

47.10 Exam retrieval and common confusions

Five distinctions prevent most errors on this topic.

1. Humanware means **people**. It is neither hardware nor software; do not confuse it with firmware (which is software stored in hardware).
2. **IT governance** is governance *of* IT; **e-governance** is IT *in* governance. Do not swap them.
3. **G2C** is Government to **Citizen**, not Government to Company. The business interaction is G2B.
4. **Authentication** proves identity; **authorization** grants permission. Authentication comes first.
5. An **electronic record** is digital information. A printed paper file is a physical record, not an electronic one.

Example 47.10.1. *An item asks which component of a computer system consists of the people who operate and manage it. Hardware is the machinery and software is the set of instructions, so both are wrong. The correct answer is humanware. If the stem instead asks what improves transparency and accountability in government, the answer is the role of IT in governance, and if it asks what verifies a user's identity, the answer is authentication.*

Exercises

1. Define humanware and name four categories of people who form it.
2. Distinguish hardware, software and humanware, and explain why all three are needed.
3. Give three benefits of using IT in governance.
4. Distinguish IT governance from e-governance.
5. What are the four types of e-governance interaction, and what does each connect?

6. Define an electronic record and give two examples.
7. Distinguish authentication from authorization, and name the three standard authentication factors.

Solutions

1. Humanware (also called liveware or peopleware) is the human element of a computer system. Four categories are users (end users), operators, administrators, and programmers (with IT support and technicians).
2. Hardware is the physical machine, software is the set of programs, and humanware is the people. Hardware without software does nothing, and both need people to design, run and use them, so all three are required.
3. Any three of: transparency, accountability, efficiency, convenience, lower cost, reduced corruption, and citizen-centric service.
4. IT governance is the direction and control of an organisation's own IT resources — governance of IT. E-governance is the use of IT by government to deliver public services — IT in governance.
5. G2C (Government to Citizen) serves the public; G2G (Government to Government) links departments; G2B (Government to Business) links companies; G2E (Government to Employee) links government staff.
6. An electronic record is information created, stored and retrieved in electronic form. Examples include an e-document, an e-file, a scanned paper or a database entry.
7. Authentication verifies identity (“who are you?”); authorization grants or denies permission (“what may you do?”). The three factors are something you know, something you have, and something you are.